

Date
14/9/23
Tuesday

CSN0315 - Data structure

Assignment - CO1 - AT1

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Q1) C program to count odd and even elements in an array

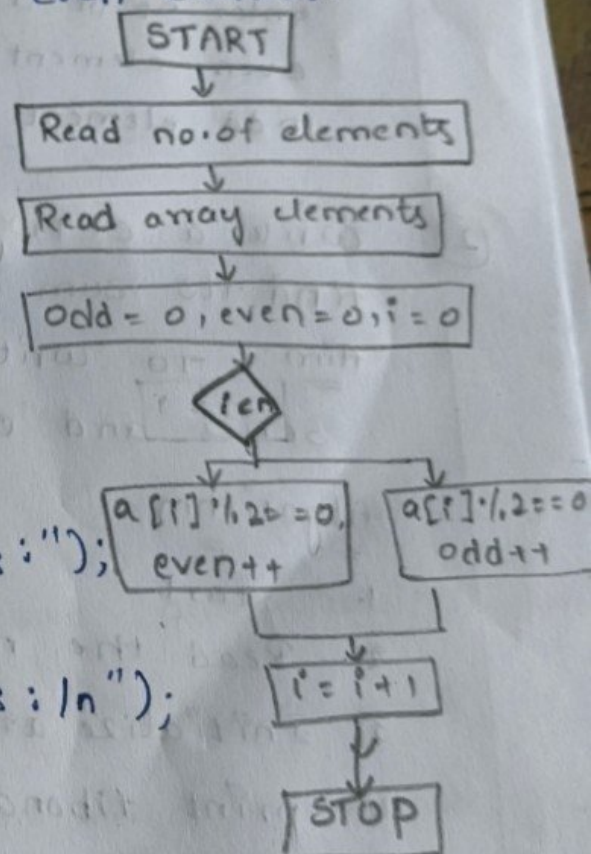
Aim: To write a C program to count the number of odd and even elements (n)

Algorithm

1. start
2. Read the number of elements (n)
3. Read the array element
4. initialize odd = 0 and even = 0
5. check each element
6. Display the no. of odd and even element.
7. stop.

Program

```
#include <stdio.h>
int main ()
{
    int n, i, a[100];
    int odd = 0, even = 0;
    printf ("enter no. of elements:");
    scanf ("%d", &n);
    printf ("enter array elements: \n");
    for (i = 0; i < n; i++)
    {
        scanf ("%d", &a[i]);
    }
```




```
for (i=0; i<n; i++)
```

```
{
```

```
if (a[i] % 2 == 0)
```

```
    even ++;
```

```
else
```

```
    odd ++;
```

```
}
```

```
printf ("even elements = %d\n", even);
```

```
printf ("odd elements = %d\n", odd);
```

```
return 0;
```

```
}
```

output:

enter no. of elements = 5

enter no. of array element = 10 15 8 7 20

even element = 3

odd element = 2

Program:-
#include

int m

{

int n

int

in

pr

②

Write a C-program to print fibonacci series and find its sum of series.

Aim:- To write a C program to print the fibonacci series and calculate its sum.

Algorithm:-

1. start

2. Read the no. of term

3. Initialize a=0, b=1, sum=0

4. print fibonacci term

5. Add each term to sum

6. Display the total sum

7. stop.

Program:-

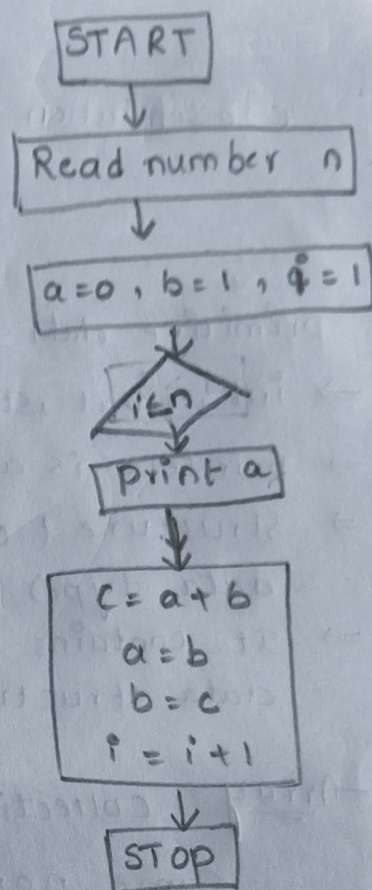
```
#include <stdio.h>
int main()
{
    int n, i;
    int a = 0, b = 1, c;
    int sum = 0;
    print ("enter no. of terms:");
    scanf ("%d", &n);
    printf ("fibonacci series : \n");
    for (i = 1; i <= n; i++)
    {
        print ("%d", a);
        sum = sum + a;
        c = a + b;
        a = b;
        b = c;
    }
    print ("sum = %d", sum);
    return 0;
}
```

output

enter no. of elements = 7

fibonacci series : 0 1 1 2 3 5 8

Sum = 20



③

concept map for data structure and classification

Defination:- A data structure is a method of organizing the data from the memory so that it can be accessed and modified efficiently.

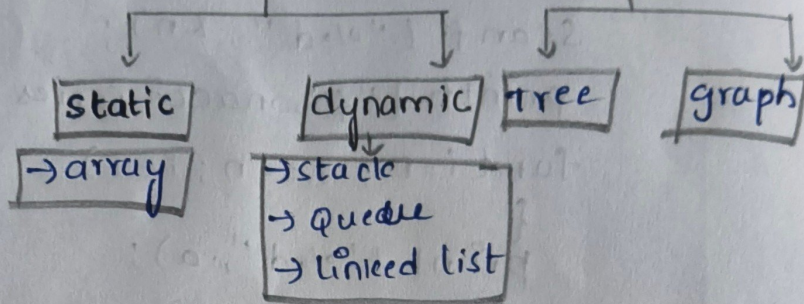
Classification of data structure

primitive data structure

- ⇒ int, float, string, char
- ⇒ pointers is a variable
- ⇒ structure (collect of data type)
- ⇒ It contains basic data structure

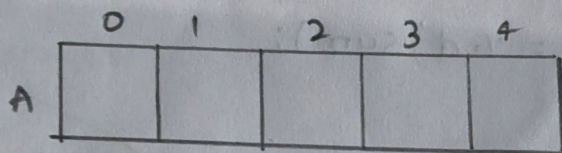
non-primitive data structure

→ **Linear** and **non-linear**



Array:- Collection of location in memory under the same name and same data type is called array.

Ex:-



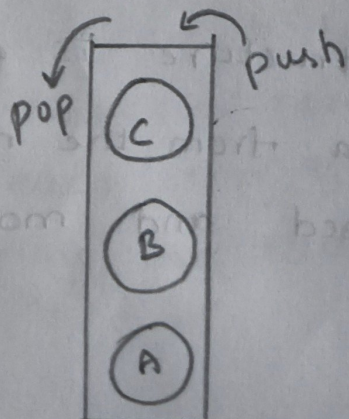
Array in data structure

Stack:-

last in first out (The boundary of data structure is fixed)

non-stack:-

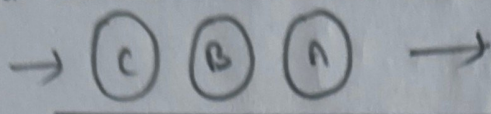
no boundary & and memory locations are not fixed.



stack of data structure (LIFO)

Queue:

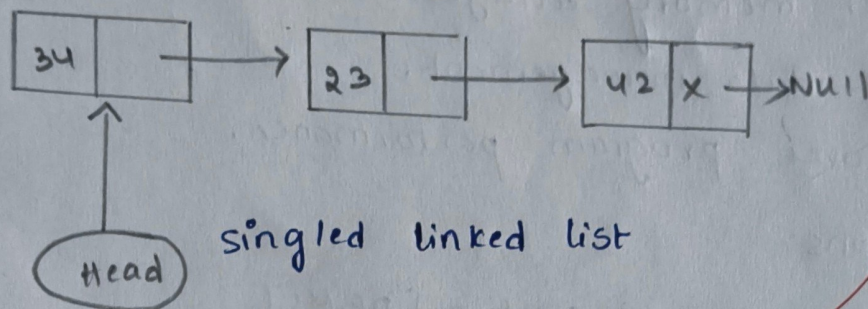
First in first order or mannan o.
enqueue dequeue



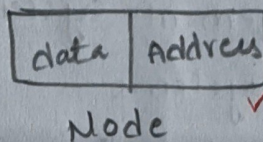
Queue data Structure (FIFO)

Linked List:

Connected to one data to another data
with the help of address.



singled linked list



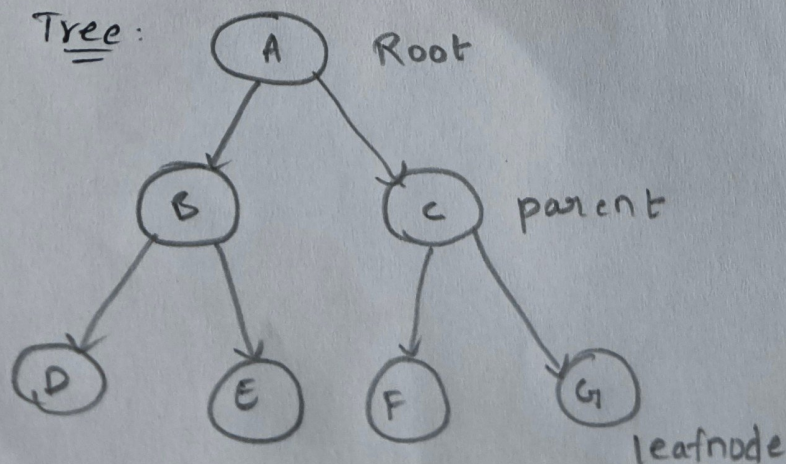
Non-linear data structure:-

⇒ It is a non-sequential order.

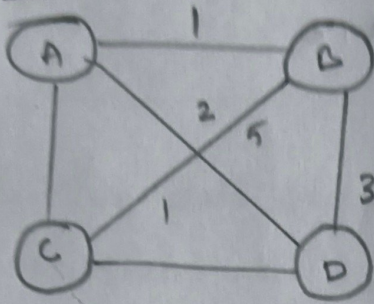
⇒ These are Classified into 2 types.

1. Tree data structure
2. Graph data structure.

Tree:



Graph:-



$$V = \{v \in E\}$$

$$V = \{A, B, C, D\}$$

$$E = \{AB, AC, BD, CD, AD\}$$

Advantages:-

- efficient storage of data
- Faster searching and sorting.
- Better memory utilization.
- Easy data management.
- improves program performance.

Applications:-

- ⇒ Data management system [DBMS]
- ⇒ Operating systems
- ⇒ computer network
- ⇒ Artificial intelligence
- ⇒ software development.